

Phase 3:Project Design Phase

System & UI Design Document

Role	Name
Member	Boora Lakshmi

System Architecture Flow

The system follows a standard client-server Machine Learning architecture, as illustrated below:

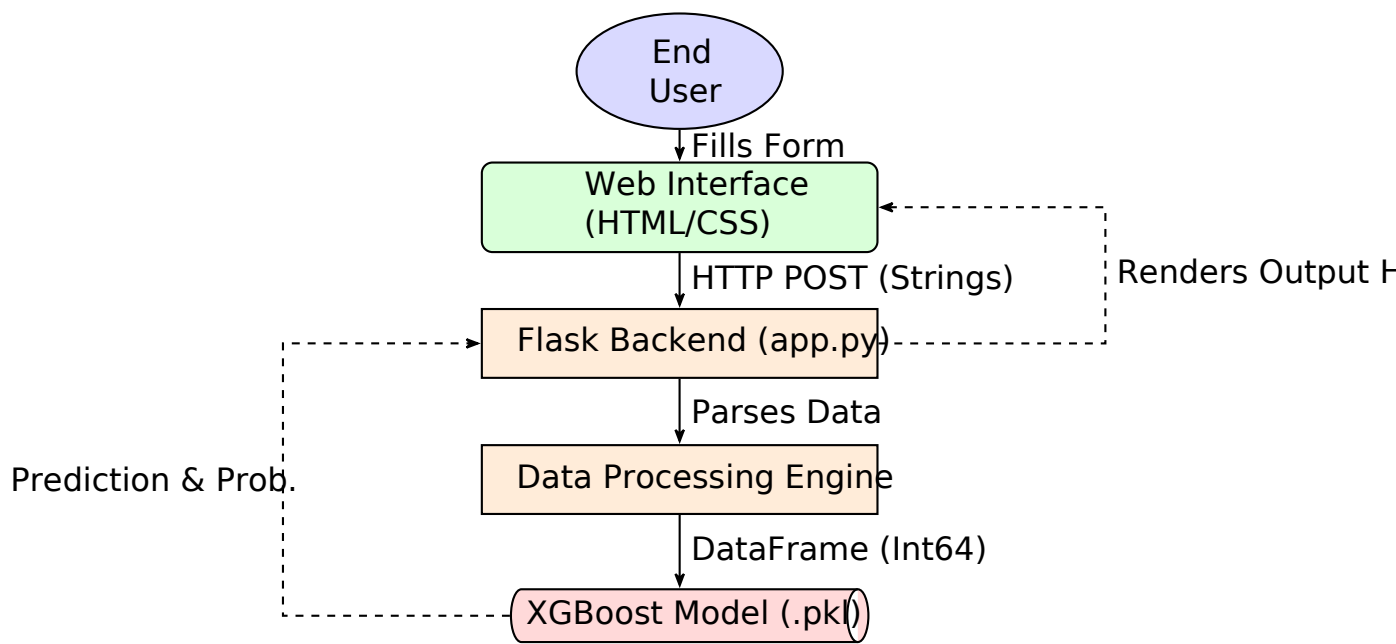


Figure 1SmartLender System Architecture Flow

Data Flow Logic

When user data arrives via HTTP POST, it is in string format. That Flask app executes a strict sequence of transformations to ensure XGBoost receives the exact int64 format it was trained on, using model.columns.pkl to prevent feature mismatch errors.

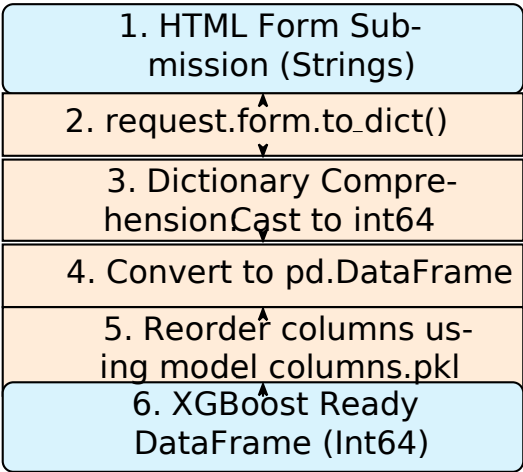


Figure 2:Data Preprocessing and Transformation Flow

Entity-Relationship (ER) Diagram

The dataset follows a single-entity structure, all features belong to one unified loan application record. The diagram below illustrates the LOAN APPLICATION entity and its grouped attributes.

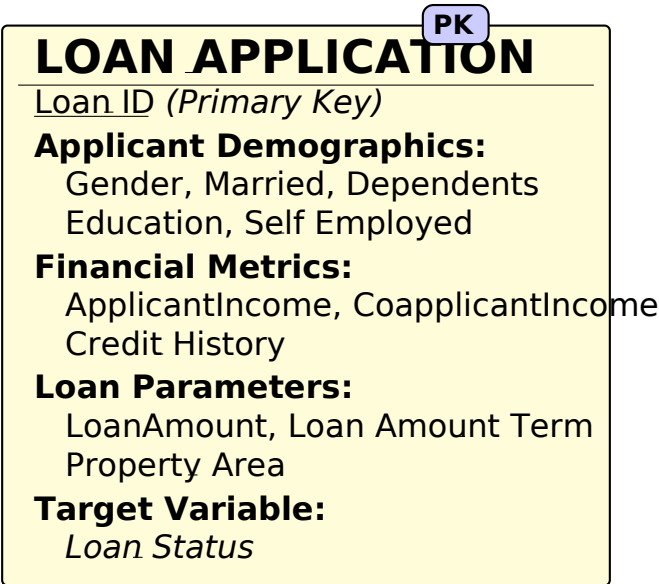


Figure 3:Entity-Relationship Diagram for Loan Dataset

User Interface Mockups

- ❖ **Home Page (home.html)** A clean landing page displaying project details, tech stack used, and a prominent green “Start Application” button that routes to the form.
- ❖ **Input Page (input.html)** A structured HTML form containing dropdowns for categorical variables (mapped to integer encodings, Male=1, Female=0) and number inputs for continuous variables like ApplicantIncome and LoanAmount.
- ❖ **Output Page (output.html)** A focused result page displaying the loan status (APPROVED in green text / REJECTED in red text) and the model’s confidence percentage.